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Structural mechanism of carbon mineralization in mechano-chemically alkali activated cementitious materials from granu-lated slag

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Accelerated carbonation and structural transformation of blast furnace slag by mechanochemical alkali-activation

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Abstract

Alternative cements and production routes are necessary to offset the considerable global CO_2 emissions of Portland cement production. The combination of alkali-activation and mechanochemical milling in a CO_2 rich atmosphere is a promising green direction for synthesizing cementitious material as it upcycles hazardous material (slag) while capturing wt% of CO_2 during synthesis. We investigate the resulting structural transformations incurred during synthesis and hydration using a suite of characterization techniques including solid-state ^{27}Al, ^{29}Si, and ^{13}C NMR. The local aluminosilicate network structure of the processed clinker is best described by a melilite-type structure. Upon hydration, the network polymerizes to form a calcium, sodium aluminosilicate hydrate gel. The synthesis route also creates various metastable carbonates and bicarbonates from captured CO_2 and alkali-additives that transform into stable carbonate phases like calcite, aragonite, and gaylussite, after hydration. This indicates accelerated carbonation reactions occur during clinker production and demonstrates novelty as a green cement technology.

1. Introduction

Production of Portland cement is estimated to account for 5% of the world's greenhouse gas emissions [1]. Most of these emissions come from the high temperature (900-1100°C) calcining process to convert $CaCO_3$ into CaO. Because of the extremely large quantities of cement produced every year, on the order of billions of metric tons, finding pathways to reduce CO_2 emissions is one of the most important research challenges for the cement industry. One pathway for CO_2 reduction involves blending Portland cement with hydraulic supplementary cementitious materials (SCM) that do not generate CO_2 during production. These SCM additives can be readily implemented by pre-existing cement production facilities although they may impact the resulting strength and composition/microstructure of the hydration product [2]. Another CO_2 reduction strategy is sequestering CO_2 during production or while curing cement to offset the net emissions. Portland cement naturally reacts with atmospheric CO_2 slowly over its lifetime. The natural carbonation reaction occurs by striping Ca ions from the pore solution to form calcium carbonate which has two detrimental effects: neutralization of the high pH environment, leading to corrosion of embedded reinforcing steel, and crazing from the high molar volume of precipitated calcium carbonate [3]. However, carbonation reactions at early curing stages have been shown to have favorable effects where the carbonation products are integrated into the binding matrix and can increase the compressive strength of the cement [4,5]. Sequestration strategies revolve around accelerating the carbonation reaction at early stages of curing by injecting pressurized CO_2 or elevating the temperature to enhance CO_2 diffusivity in the pore solution [6–9]. These approaches require additional equipment and high capital investments that impedes widespread implementation.

Adoption of alternative cement chemistry that does not require calcining the reactants for production can further diminish the CO_2 footprint. Alkali-activated materials (AAM) are a promising cement alternative [2] that are created by exposing anhydrous aluminosilicates to an alkali-rich solution. Sources for the anhydrous aluminosilicate material come from calcium and magnesium-rich industrial residues like fly ash, granulated blast furnace slag (GBFS), and mine waste [10] which are otherwise used in low-value applications or classified as hazardous waste materials. Upcycling these waste materials to be the source for cementitious materials prevents their deleterious caustic effects on the environment and potential pollution in waste streams [11]. The relatively reduced CO_2 footprint and upcycling aspect makes AAM a very promising green technology in the cement application area [12]. However, AAM needs to be competitive with Portland cement in terms of mechanical and chemical performance as well as economic value before it can see widespread adoption. A thorough description of the microstructure and phase assemblage is critical towards understanding the mechanical and chemical properties of AAM and cementitious materials. A microstructural understanding of AAM will enable greater engineering capabilities for its use as an alternative green cement.

The microstructure of AAM shares some characteristics with Portland cement as both are an ensemble of many amorphous inorganic silicate phases of variable composition and degrees of crystallinity. The main binder is calcium-aluminosilicate-hydrate (C-A-S-H) gel whose structure is modelled by tobermorite having alternating layers of calcium ions, aluminosilicate chains, and water molecules [13]. The structure of the aluminosilicate chain is composed of two types of Si tetrahedra, pairing and bridging, where the bridging site crosslinks two pairs of the pairing tetrahedra. In terms of the Q^n nomenclature in which *n* denotes the number of bridging oxygens per Si tetrahedra, both bridging and pairing tetrahedra are Q^2 units. The C-A-S-H gel is primarily Q^2 units with varying amounts of Q^1 and Q^3 units representing chain end members and points of chain crosslinking, respectively [13–15]. Al atoms can substitute either of the Si bridging or pairing sites to promote cross-linking of the chains. An analogous nomenclature for describing the number of bridging oxygen per Al tetrahedra denoted by q^n. The quantity of each short range structural unit is determined by the concentration of Ca, Si, and Al in the source material [2]. Alkali activation incorporates alkali ions, like sodium, into the gel interlayer where they charge balance AlO_4^-. These CaO-Na_2O-Al_2O_3-SiO_2-H gels (C-N-A-S-H) have been described in terms of a mixture of cross-linked and non-cross-linked tobermorite-based structure [16,17]. The microstructure of precursor waste AAM materials typically are composed of short aluminosilicate structural units, Q^1, that are further depolymerized when exposed to the alkaline solution. After saturation of the solution, the aluminosilicate gel precipitates to form short molecular units of Q^1, and Q^2/q^2 units that polymerize over time to form longer C(N)-A-S-H gel network composed of Q^2, Q^3, and q^3 units [10].

The high concentration of Ca ions in AAM also make them appealing for carbonation reactions to sequester CO_2 although under regular curing conditions these carbonation reactions are diffusion limited and become increasingly slow as passivating carbonate layers builds up [18,19]. Mechanochemically milling the materials can increase the rate of carbonation reaction by continually breaking the carbonate layer and exposing fresh, unreacted surfaces during milling. Prior studies have demonstrated this accelerated carbonation effect by mechanochemically milling silicate minerals, GBFS, and fly ash [20–24]. Combining AAM with mechanochemical milling is a promising strategy to mitigate CO_2 production of cementitious materials by converting previously waste materials into stable cement displaying accelerated carbonation products. This mechanochemical milling alkali activation process (MCAA) is performed at room temperature with relatively little energy input and can be carried out with pre-existing equipment at cement production facilities. Prior investigations into mechanochemically milling alkali-activated aluminosilicates have shown the resulting cement product meets performance specifications required for general use cement and can have higher compressive strength and acid resistance relative to Portland cement [25,26]. Milling in CO_2 atmospheres rather than in air has shown to increase the CO_2 uptake although can have mixed results in terms of impacting compressive strength development [20,27,28]. The initial alkaline source and quantity before milling likely plays a major role in determining the resulting properties of the hydrated cementitious product. The MCAA process has been recently demonstrated at a pilot-scale and the resulting hydraulic cement was found to be compatible with conventional construction practices in the field [29]. However, a thorough investigation of the microstructure incurred by the MCAA process is still lacking. Mechanochemical milling of AAM will dramatically change the short-range structure by increasing amorphization and needs to be further explored to understand the impact on gel network structure and carbonation pathways.

Here, we report our findings on the structural transformation of GBFS into a hydraulic cementitious material by a simultaneous mechanochemical milling alkali activation process. Structural analysis on the precursor GBFS, reacted material after the MCAA process, and the MCAA material after hydration is performed to track the structural transformation and state of carbonation. Structural analysis is performed by X-ray diffraction (XRD) for identifying crystalline phases, while 1- and 2D ^{27}Al and ^{29}Si solid state nuclear magnetic resonance (NMR) experiments elucidate the amorphous network structure. Complementary structural information is obtained with Fourier transform infrared spectroscopy (FTIR). The state of carbonation is determined with ^{13}C solid-state NMR and by analyzing the thermal decomposition using thermogravimetric analysis (TGA) and FTIR of the outgassing products. We find the MCAA process transforms the GBFS into a cementitious material containing significant amounts of metastable carbonates that then transform into stable carbonate phases upon hydration along with polymerizing the aluminosilicate network. The MCAA process is a promising direction for producing green cement by upcycling waste slag into a cementitious material while also capturing wt% of CO_2 at room temperature in a single processing step.

2. Material and Methods

2.1 The mechanochemical alkali activation (MCAA) process

The GBFS sample was analyzed as received from the source facility (Lafarge). The mechanochemical alkali-activated powder is a mixture of 90 wt% GBFS and 10 wt% Na_2CO_3, 4 wt% H_2O was then added to the mixture and loaded into a steel rotary ball mill (jar volume 12.44 L). The milling media were 25-50 mm diameter steel balls with a ball to raw material ratio of 36:1. The CO_2 gas flow into the jar was 10 standard cubic feet per hour (SCFH) while milling the mixture for 2 hours at 84 rpm. The resulting powder, referred to hereon as the MCAA powder, was formulated into a hydraulic cement using a 2.5:1 powder to water ratio. The samples were left to cure for a minimum of 19 days at room temperature and in air then transferred to a closed container vial for storage up to 8 months prior to analysis. A fresh paste sample was made for scanning electron microscopy where a 2.5:1 hydraulic cement to water formulation was mixed and allowed to cure overnight within a vacuum desiccator.

2.2 Multinuclear Solid-State Nuclear Magnetic Resonance Spectroscopy

Triple quantum magic angle spinning (3QMAS) ^{27}Al NMR was collected on a 600 MHz Avance III spectrometer operating at a Larmor frequency of 156.38 MHz. The samples were spun at 30 kHz in 2.5mm ZrO_2 rotors. A split t_1 3QMAS with double frequency sweep (DFS) sequence was used to enhance the sensitivity of the relatively weak ^{27}Al signal and increase resolution. A RF field strength of 83 kHz was used for the excitation pulse, having a pulse length of 3 µs followed by the DFS conversion pulse length of 11.1 µs. A RF field strength of 20.8 kHz was used for the 180° selective pulse (24 µs). For 2D acquisition, the rotor synchronized sequence was incremented through 32 t_1 steps collecting 4128 transients per t_1 increment with a recycle delay of 0.5 s. The single-pulse and cross-polarization solid-state MAS NMR experiments for ^{29}Si and ^{13}C were collected on a 400 MHz Avance III spectrometer operating at Larmor frequencies 79.44 MHz and 100.54 MHz, respectively. The samples were packed in 4 mm ZrO_2 rotors and spun at 10 kHz. Single pulse ^{29}Si spectra were collected with a $\pi/2$ pulse length of 4.5 µs (B_1 field = 55.5 kHz) and single pulse ^{13}C spectra were collected with a $\pi/2$ pulse length of 4.875 µs (B_1 field = 51.28 kHz). Between 1 000 and 10 000 transients were collected for each spectrum with recycle delays of 15 s and 60 s for ^{29}Si and ^{13}C experiments, respectively. The ^{29}Si and ^{13}C cross polarization (CP) MAS experiments were collected with a contact time of 2 ms and 4 ms, respectively, and used spinal64 for decoupling. Kaolinite and α-glycine were used as standards for setting up the CP experiments.

The inherent structural disorder present in slag and cementitious materials severely broadens ^{29}Si MAS NMR line shapes and prevents satisfactory site assignment due to limited peak resolution. The 2D magic angle turning phase adjusted sideband separation (MATPASS)/CPMG experiment can resolve this problem by separating the chemical shift anisotropy (CSA) from the isotropic contribution. This isotropic-CSA separation serves two purposes, producing a 'CSA-free' isotropic line shape that has improved resolution compared to conventional spinning speeds and because the CSA information is retained in the indirect dimension it can be correlated with the isotropic direct dimension on a point-by-point basis. Unique sites with overlapping resonances in the isotropic dimension can then be identified based on their CSA parameters. The MATPASS/CPMG experiment is a rotor synchronized pulse sequence with five π pulses following inter-pulse delays outlined by Hung et al. [30] The π pulse length was 8.06 μs with a spin speed of 1.567 kHz and for 2D acquisition 32 hypercomplex t_1 points were collected with 840 transients per t_1 point and 23 echoes per transient with a recycle delay of 15 s. The method of STATES [31] was used for acquisition of the hypercomplex data and applied to the phases of the receiver and CPMG pulses. The 2D MATPASS/CPMG dataset was processed using a custom Python script that follows typical MATPASS/CPMG processing steps in addition to fitting the T_2 decays from the CPMG echo train and adjusting their spectral components through CPMG reconstruction [32,33]. The implementation of the CPMG reconstruction processing with the MATPASS/CPMG pulse sequence will be discussed in a forthcoming publication.

The ^{27}Al, ^{29}Si, and ^{13}C chemical shifts were externally referenced to the isotropic chemical shift of 0.1M $AlCl_3$ (δ_{iso}= 0 ppm), kaolinite (δ_{iso}= -92.0 ppm), and α-glycine (δ_{iso}= -176.5 ppm), respectively. Processing of the 3QMAS spectra were carried out using Bruker TopSpin software. Deconvolution of the ^{27}Al, ^{29}Si, and ^{13}C line shapes and analysis of the ^{27}Al quadrupolar coupling constants was carried out using the software package Dmfit [34]. The principal components of the chemical shift tensor (δ_{xx}, δ_{yy}, δ_{zz}) are expressed in accordance with the Haeberlen convention by the isotropic chemical shift δ_{iso}, magnitude of anisotropy Δ, and asymmetry parameter η. These parameters are defined as:

$$\delta_{iso} = \frac{1}{3}\left(\delta_{xx} + \delta_{yy} + \delta_{zz}\right); \Delta = \delta_{zz} - \delta_{iso}; \eta = \frac{\delta_{yy} - \delta_{xx}}{\delta_{zz} - \delta_{iso}}$$

where the principal components are ordered by $|\delta_{zz} - \delta_{iso}| \geq |\delta_{xx} - \delta_{iso}| \geq |\delta_{yy} - \delta_{iso}|$.

The solid-state NMR measurements were performed on the paste sample four months after hydration, this is a sufficiently long enough curing time for the hydraulic reaction to near completion. Thus, the solid-state NMR measurements reflect the final state of the cured cement.

2.3 Multi-analytical Characterization Techniques

In addition to the solid-state NMR spectroscopy, the precursor slag, MCAA powder and paste were characterized with a variety of techniques: powder X-Ray diffraction (XRD), scanning electron microscopy (SEM), Fourier transform infrared spectroscopy (FTIR), thermogravimetric analysis (TGA), variable temperature IR spectral measurements, and total inorganic carbon content (TIC). XRD patterns were collected on a Bruker AXS D8 Advance diffractometer using Cu Kα radiation between 12-56° 2θ using a step size of 0.015° and 3 seconds per step. A knife edge filter was used to reduce the background reflections at low angles of 2θ. Carbon coated powdered samples were analyzed with scanning electron microscopy by detection of secondary electrons and energy dispersive X-ray spectroscopy (EDS). FTIR spectroscopy was collected on samples pressed into pellets made with KBr using a Jasco FTIR 4600 spectrometer from 400 to 4000 cm^{-1} with a spectral resolution of 4 cm^{-1} and each spectrum is the average of 32 scans. TGA was collected using a Perkin Elmer TGA 400 at a heating rate of 15°C/min from 50°C to 900°C under flowing N_2 (20 ml/min). The TIC measurements were collected in triplicate by UIC Inc. The full details of the experimental conditions for the variable temperature IR spectral measurements and schematic of the experimental setup are provided in the supporting information. Before the sample is heated at a rate of 10°C/min, the cell is pressurized up to 1 atm with nitrogen gas and the reference spectrum, $I_{ref}(\nu)$, is recorded using the FTIR. The sample spectrum, $I_{sample}(\nu)$, is acquired as a function of temperature during heating of the concrete sample. The intensity ratio, I_{Sample}/I_{ref}, is calculated to determine the absorption spectrum from the Beer-Lambert law, $\alpha_\nu = -ln(I_{Sample}/I_{ref})_\nu$, where $\alpha_\nu$is the absorbance at wavenumber ν.

3. Results and Discussion

3.1 Crystalline Microstructure via XRD and SEM.

Due to the complex nature of the transformation of slag to hydraulic cement, we have used a multi-analytical approach to fully describe these processes. Insight into the crystalline components and microstructure are derived from XRD and SEM measurements (Fig. 1). All XRD powder patterns (Fig. 1a) are dominated by an amorphous background signal with minor amount of crystalline signal, reflecting the largely amorphous character of these materials. The precursor granulated slag shows minor reflections from calcite ($CaCO_3$ PDF 00-066-0867) and merwinite ($Ca_3Mg(SiO_4)_2$ PDF 00-035-0591) which have been observed in other precursor slags [35–37]. After the MCAA process, those reflections are replaced by new reflections from trona ($Na_3H(CO_3)_2$•$2H_2O$ PDF 00-029-1447) and nahcolite ($NaHCO_3$ PDF 00-015-0700). Hydration of the MCAA powder results in dissolution of the bicarbonate phases and subsequent precipitation of calcite ($CaCO_3$ PDF 00-005-05686), aragonite ($CaCO_3$ PDF 00-041-1475), and gaylussite ($Na_2Ca(CO_3)_2(H_2O)_5$ PDF 04-010-3621) due to hydration reactions. These reactions have been observed previously in alkali activated mixtures of slag and sodium carbonate where cation exchange reactions produce gaylussite [38,39]. The formation of many distinct carbonate species in the hydrated material is strong evidence of the propensity of MCAA process for accelerated carbonation. SEM analysis provides further insight into the development of crystalline phases. Micrographs of the precursor slag (Fig. 1b) reveal micron scale whisker formations of calcite on the surface of the slag. The MCAA process destroys the surface morphology resulting in featureless grains (Fig. 1c). After hydration (Fig. 1d), strip like features are present on the surface of the bulk material and grow in density with prolonged curing times (Fig. 1e). These new needles are identified as primarily carbonate phases (calcite, aragonite, and gaylussite) given no other notable crystalline phases are observed. Similar studies on alkali activated slag have noted similar needle like surface morphology occurring after hydration that is attributed to precipitation of calcite and trona [40]. EDS measurements reveal that the composition of the slag did not appreciably change outside of the addition of sodium carbonate during the MCAA process. The composition of the slag has been previously reported [20,28] and is close to that of merwinite which is an impurity phase observed within the slag. The composition of the MCAA powder and paste are similar and indicate the Ca/Mg ratio is ~6, Ca/Si ratio is 1.46, Al/Si ratio is 0.3, Ca+Mg/Na ratio is ~ 3.5, and the total modifier (Ca+Mg+Na) to network former (Al+Si) ratio is ~1.6. These ratios provide important constraints and considerations for interpreting subsequent data.

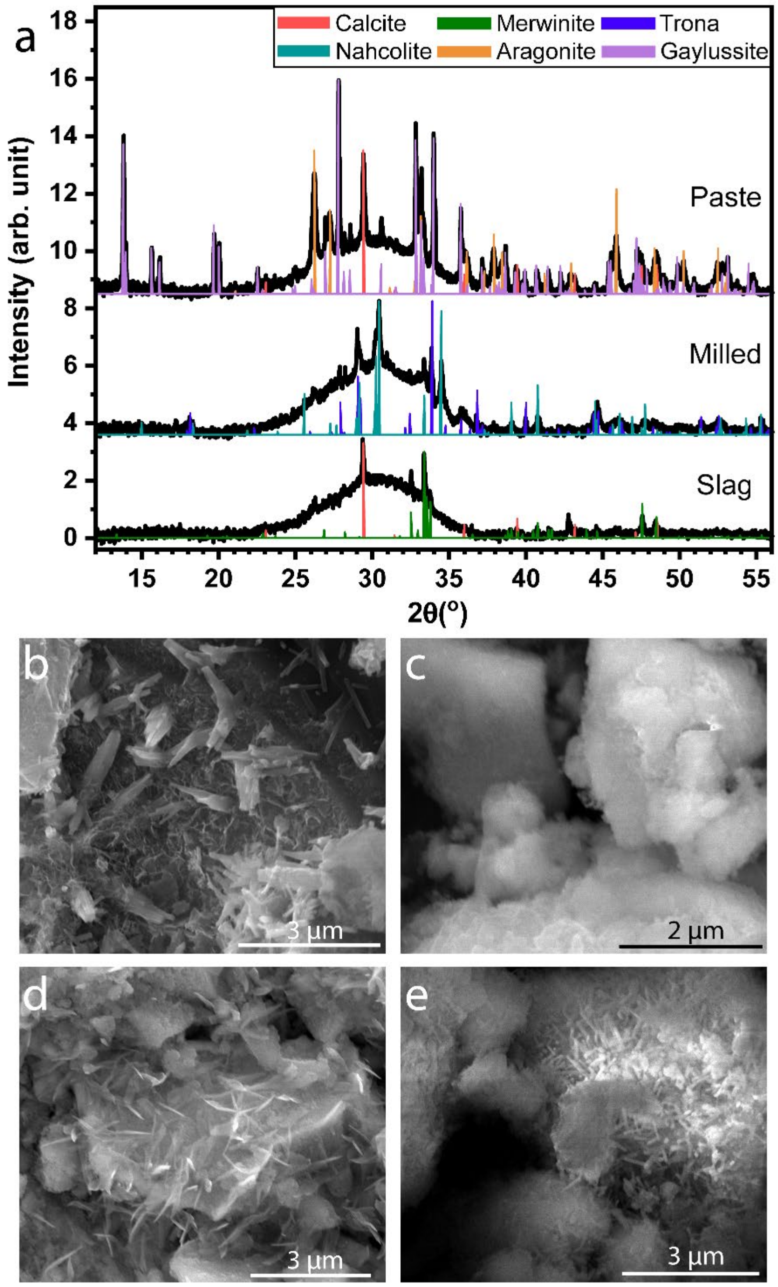

Figure 1 a. XRD patterns of slag, MCAA powder, and MCAA paste with reflections from mineral phases denoted by color lines. SEM micrographs of b. granulated slag, c. MCAA powder, d. overnight cured MCAA paste, e. long cured MCAA paste.

3.2 Structure of amorphous components from 1- and 2D solid-state NMR.

The ^{27}Al and ^{29}Si NMR data collected on these samples reveal detailed information into their amorphous microstructure. Typical ^{27}Al and ^{29}Si line shapes of AAM cement formulations contain many narrow and broad resonances corresponding to the numerous constituent crystalline and amorphous phases [17,41,42]. However, the MCAA NMR line shapes are broader in comparison indicating these materials are much more disordered and contain a higher fraction of amorphous phases. Despite the broadening, 3QMAS ^{27}Al NMR can be used to improve the spectral resolution [43] and assign peaks based on their quadrupolar coupling constants (C_Q). The ^{27}Al isotropic projections of 3QMAS NMR are shown in figure 2 with the corresponding deconvolution of the Al environments.

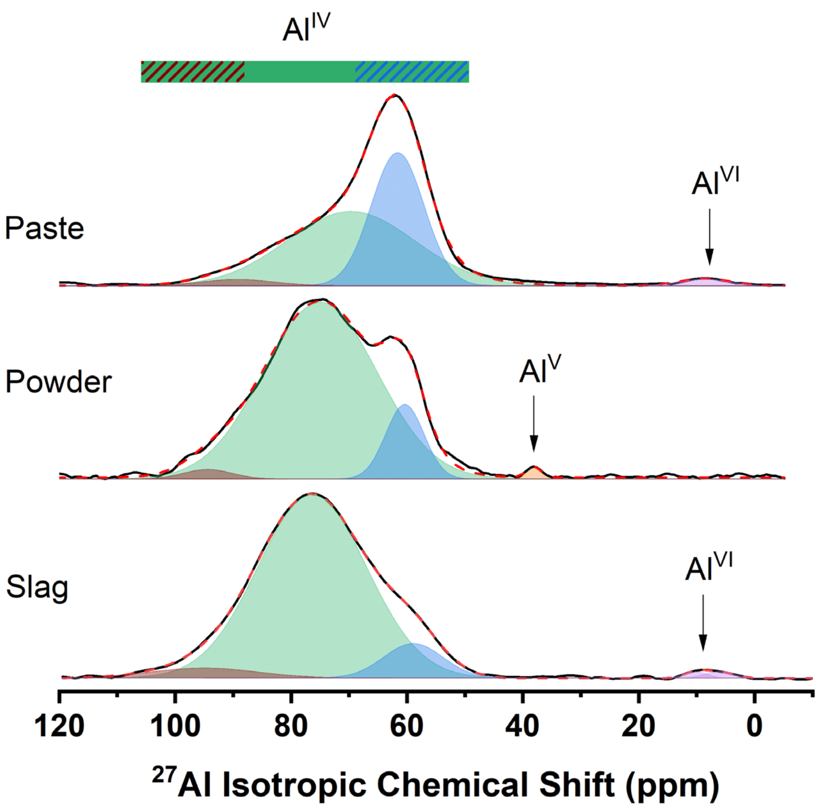

Figure 2 ^{27}Al 3QMAS isotropic projection of slag, MCAA powder, and MCAA paste samples. Corresponding deconvolution parameters are in Table S1. The dashed and solid color bar denotes the chemical shift range of the three Al^{IV} sites.

The isotropic ^{27}Al line shapes for all the samples display significant chemical broadening due to the inherent disorder of the aluminosilicate network. Nearly all Al are in tetrahedrally coordinated environments (Al^{IV}) across the series with minor exceptions (<2%) of octahedral Al (Al^{VI}) and five-coordinated Al (Al^{V}). There are three distinct Al^{IV} sites that can be assigned based on their respective isotropic chemical shift and C_Q (Table S1). It should be noted, a Czjzeck distribution is required to adequately fit these line shape indicating the local environment of Al is structurally disordered [44]. The average quadrupolar coupling constants of the three Al^{IV} environments are unique (8.7 MHz at δ^{F1} = 95 ppm, 6.7 MHz at δ^{F1} = 76 ppm, and 2.8 MHz at δ^{F1} = 61 ppm) and remain relatively unchanged throughout the MCAA and hydration process. It is notable that all of these Al C_Q values are significantly larger than those found in C-S-H/C-A-S-H gels, tobermorite, and aluminate hydrate environments (typically less than 2 MHz) [13,45,46]. This observation, along with the absence of significant concentrations of octahedral Al, strongly suggests these materials do not form typical C-A-S-H-like gels commonly observed in alkali-activated cementitious materials, rather an alternative gel structure is formed that bares some similarity to N-A-S-H [41].

Insight into the slag and MCAA gel microstructure can be gained by comparing the δ_{iso} and C_Q of the Al^{IV} environments (Table S1) to those in minerals. For the slag and the MCAA powder, the δ_{iso} and C_Q of the primary tetrahedral peak, Al^{IV}_1, falls neatly in the range observed in the mineral gehlenite for the T_1 site (δ_{iso} = 79.2-70.2 ppm and C_Q = 5.82-7.59 MHz) [47]. The δ_{iso} and C_Q of the minor Al^{IV}_2 site also agrees with the range found for Al in the T_2 site in gehlenite (δ_{iso} = 87-90 ppm and C_Q = 8.3-10.7 MHz). Gehlenite, ($Ca_2Al_2SiO_7$), is a layered structure of alternating aluminosilicate and Ca ion planes. Within the aluminosilicate plane, there are two tetrahedral sites that display partial Al/Si ordering the paired tetrahedra, T_2, and bridging tetrahedra, T_1. The paired tetrahedra are pseudo isolated dimer pairs of tetrahedra that have one non-bridging oxygen out of the aluminosilicate plane and can be occupied by either Al or Si. Two of the corners of the T_2 tetrahedra are bonded to the bridging tetrahedral species T_1 which are exclusively occupied by Al and share all four corners with neighboring T_2 sites. As a member of the melilite mineral group, gehlenite forms a solid solution with the mineral akermanite, $Ca_2MgSi_2O_7$, where Mg exclusively resides on the T_1 sites and the rest of the structure remains the same. The third Al^{IV} peak, having highest concentration in the MCAA paste, has a notably smaller C_Q (2.8 MHz) than the other tetrahedral sites suggesting it is charge balanced by lower valence cations like H^+ or Na^+ [48,49]. The chemical shift and C_Q are in good agreement with the cross-linking q^3 and q^4 species found in C-A-S-H and N-A-S-H type gels, respectively (δ_{iso} = 60 ppm, C_Q = 4.1 MHz) [41].

The δ_{iso} and C_Q of the tetrahedral sites do not change much after the MCAA process and hydration (Table S1), however the relative intensity of these peaks changes considerably and reflects an evolving microstructure. The granulated slag is predominantly composed of a gehlenite-type (T_1) bridging tetrahedral Al (Al^{IV}_1) environment centered at 76 ppm with minor amounts of the gehlenite-type (T_2) paired tetrahedra (Al^{IV}_2) appearing as a shoulder at 93 ppm and q^3 (Al^{IV}_3) in slightly higher concentration as a shoulder at ~60 ppm. After the MCAA process, the concentration of Al^{IV}_1 remains the same however the intensity of Al^{IV}_3 peak increases at the expense of the Al^{IV}_2 peak. Further changes are observed after hydration as the concentration of the Al^{IV}_3 site approaches that of the Al^{IV}_1 site while the intensity of the Al^{IV}_2 site is nearly unchanged. While small in overall intensity, it is interesting to note that the Al^{VI} sites disappear and Al^{V} sites appear after milling, then after hydration the Al^{V} disappears as Al^{VI} reforms. This observation suggests a relationship between these two Al sites; potentially the Al^{VI} is structurally distorted during milling to form metastable Al^{V} which then reverts to Al^{VI} after hydration.

The 3QMAS ^{27}Al NMR results indicate the short-range structure of Al in the slag and MCAA powder resembles a disordered form of gehlenite or a melilite solid solution. The large concentration of gehlenite-type Al remaining after the MCAA processing suggests incomplete reaction of the slag and carries important insight into understanding the resulting aluminosilicate network. Most of the aluminosilicate network in the slag and MCAA powder is composed of paired and bridging tetrahedra with Al primarily residing as bridging tetrahedra (T_1 sites) forming a condensed melilite-type layer structure. Hydrating the MCAA powder transforms some of the T_1 sites into q^3 sites that cross-link paired and bridging silicate chains similar to those observed in mixed cross-linked/noncross-linked tobermorite-type C-N-A-S-H gel structures [41]. The resulting gel network in the MCAA paste is heterogeneous containing clusters of partially reacted melilite with corner sharing AlO_4 and SiO_4 units separated by channels of Q^1(mAl) or Q^2(mAl) SiO_4 tetrahedra like those found in tobermorite-like C-A-S-H gels [13]. This structural model and the dominance of the tetrahedral Al intensity and negligible amount of octahedral Al is consistent with the modified network model for alkaline-earth aluminosilicate glasses and melilite based glasses with large Ca/Al ratios and is a strong indication these materials are an appropriate framework to understand the microstructure of the aluminosilicate network [50,51]. Inherent to both gehlenite and melilite based glasses is breakdown of Loewenstein's rule of alumina avoidance by the formation of some Al-O-Al linkages [47,50]. The presence of gehlenite-type T_2 sites in the slag and MCAA processed materials indicates these materials are no different and have some regions that do not abide by the Al avoidance principle. Likewise, the rise of the q^3 peak after the MCAA process and hydration relative to the T_1 and T_2 sites implies other regions of the aluminosilicate network re-establish Al avoidance by forming Si-O-Al linkages that dominate the typical C-(N)-A-S-H gel structural models. This emergence of Si-O-Al linkages could be driven by the lower cation concentration as the various Ca and Na carbonate phases precipitate out of the gel during hydration (section 3.3).

Further insight into the short-range structure of the aluminosilicate network is obtained by ^{29}Si magic angle spinning (MAS) NMR, the single pulse MAS spectra are shown in Fig. 3a. Evident in the single pulse spectra line shapes is a slight Lorentzian character that has been noted before in C-A-S-H gels [13,52]. The ^{29}Si spectra all display a broad peak around -75.7 ppm and the paste has an additional peak at -88.7 ppm, these are assigned to Q^1 and Q^2 environments, respectively. The Q^1 assignment is based on the similarity of the isotropic chemical shift to the Q^1 environments in rankinite (δ_{iso}= -74.5 & -75.9 ppm) [53]. As the ^{27}Al NMR indicates some gehlenite-type silicate environments should be expected, these have δ_{iso} = -72.5 and -74.7 ppm [47] and likely contribute to the peak intensity observed in the MCAA ^{29}Si spectra. It should be noted gehlenite is a sorosilicate mineral however from the Q^n notation for silicate glasses these species would be classified as Q^3(2Al) as Si shares bridging oxygen with two neighboring Al.

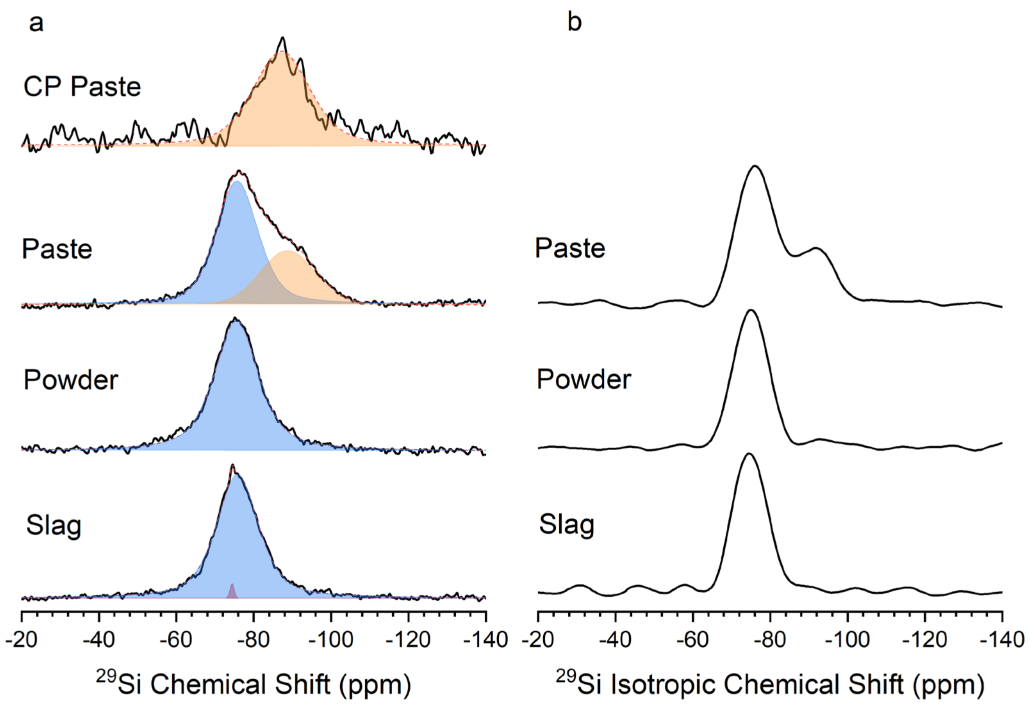

Figure 3 a. Single pulse MAS ^{29}Si spectra and {^{1}H}-^{29}Si cross-polarization spectrum of the slag, MCAA powder and paste samples and their corresponding deconvolution (see section 3.2 for details) b. ^{29}Si isotropic spectra from the MATPASS/CPMG experiment.

The additional peak at -88.7 ppm in the paste sample fits neatly within the range expected for Q^2 environments in silicates like wollastonite and the width covers the range expected for Q^2(1Al), Q^2, and Q^3(1Al) (-82, -85, -88 ppm, respectively) environments in C-(N)-A-S-H gels [41,53,54]. Within the spectra for the slag, the sharp peak at -74.3 ppm is from merwinite phase within the glass, in agreement with the XRD pattern [35]. After the MCAA process the ^{29}Si line shape remains nearly identical to the pre-milled slag suggesting the aluminosilicate network is not greatly altered and is almost entirely composed of Q^1 units. However, as will be discussed, there are significant changes observed in the MCAA powder's chemical shift anisotropy. After hydration, the line shape of the paste clearly shows the Q^2 peak indicating some of the Q^1 environments condense to form chain structures upon hydration and curing. For further insight, ^{29}Si{^{1}H} cross polarization MAS (CP/MAS) NMR reveals whether these silicate environments are near protons like hydroxyl groups or rigid structural water. The slag and MCAA powder CP spectra (not shown) do not show any signal indicating the aluminosilicate environments are largely free of hydroxyl groups. However, the paste CP spectra has signal intensity with a peak centered at -87 ppm revealing the Q^2 species contain hydroxyl groups. The occurrence of this species indicates the curing process polymerizes a portion of the Q^1 environments into hydrated Q^2 environments, which are the basic structural motif found in C-A-S-H gels.

A more comprehensive understanding of the local aluminosilicate network structure is obtained by investigating the ^{29}Si CSA of the constituent structural units. We use a 2D anisotropic-isotropic correlation technique, MATPASS/CPMG, to improve resolution of the isotropic line shape and provide information on how the CSA changes as a function of isotropic chemical shift.

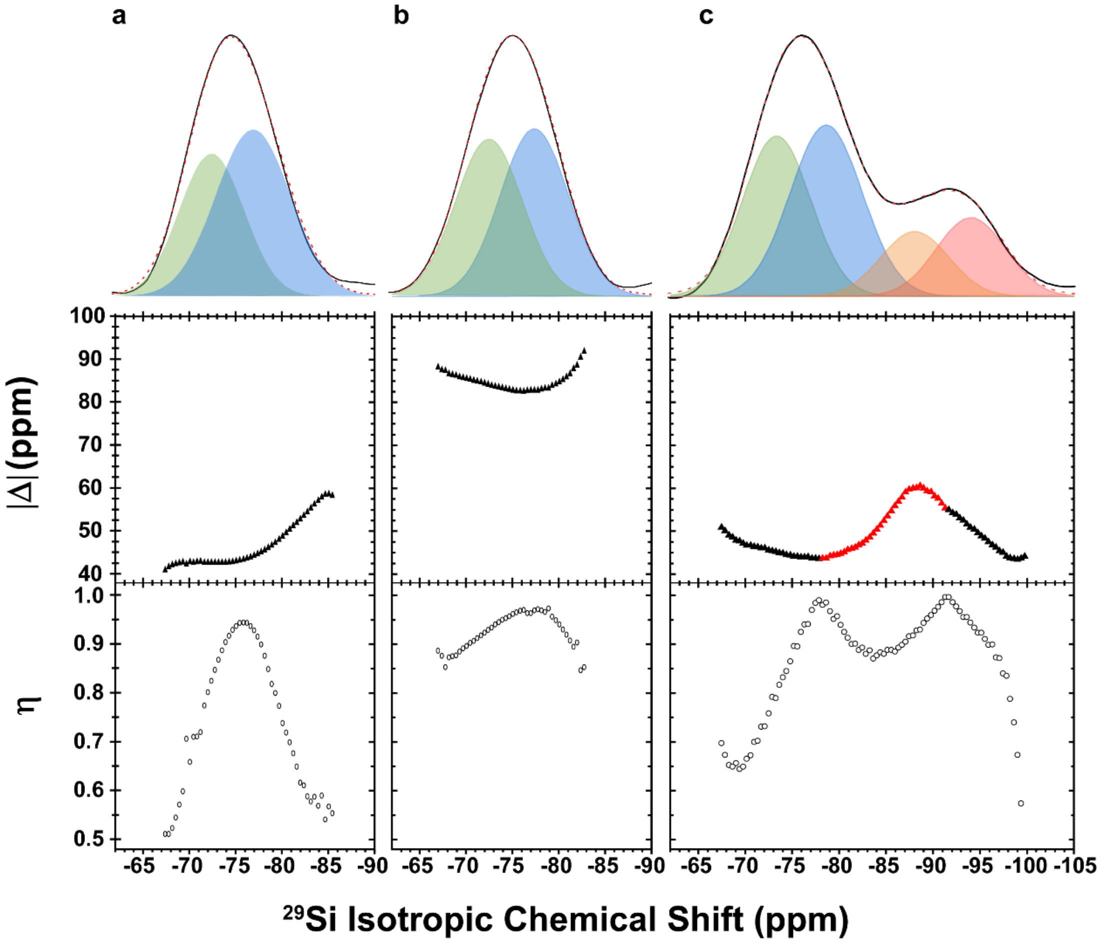

Figure 4 ^{29}Si MATPASS isotropic spectrum and deconvolution (top), CSA variation of anisotropy Δ (triangles) and asymmetry η (open circles) with isotropic chemical shift δ_{iso} (below) for the samples: a. slag, b. MCAA powder, c. MCAA paste. The red Δ symbols indicate negative Δ values.

By monitoring the CSA variation with δ_{iso} distinct chemical environments can be resolved and assigned based on their CSA parameters despite the extensive broadening within the isotropic line shape due to structural disorder. The isotropic spectra are free of any anisotropic contributions and display enhanced resolution compared to spectra collected by single pulse MAS experiments (Fig. 3b), however no additional peaks can be resolved. Analyzing the variation of the CSA parameters, anisotropy Δ and asymmetry η, with isotropic chemical shift clearly shows significant changes in the CSA within samples (Fig. 4). This variation of CSA indicates the presence distinct chemical environments and aids in deconvolution of the isotropic spectra (Fig. 4 and Table S2). Consideration of the CSA variation indicates the Q^1 peak is composed of at least two components and as well as the Q^2 peak in the paste. Overall the CSA parameter ranges for all the samples fall within typical values expected for various Q^1 and Q^2 species in silicates and C-(A)-S-H gels [53,55,56]. Within the slag, Δ increases from a plateau value of 43 ppm towards 60 ppm as δ_{iso} decreases, while concurrently η increases from 0.5 to 0.95 then decreases to 0.55. It should be noted that Q^1 environments largely have negative Δ but can have positive Δ values in C-S-H gels [56], while Q^2 environments always have positive Δ values. By nature of the convention, if two sites with opposite signs of Δ are overlapping the superposition of their sideband patterns results in an η of 1.0. The rise of η towards 1.0 in the slag likely indicates the presence of an additional chemical environment with negative Δ that is too low intensity to observe. There are many minor contributions to the CSA from additional sites due to the complexity of phases in cement chemistry however assignment of major components is defined by plateau values of CSA parameters. The two Q^1 components in the slag are assigned to gehlenite and rankinite-type environments for the lower and higher frequency peaks, respectively. These assignments are based on agreement of δ_{iso} and CSA parameters with the respective minerals gehlenite (δ_{iso} = -72.5 ppm with Δ = -44.3 ppm and η =0.45 and δ_{iso} =-74.7 ppm with Δ = -48.3 ppm and η = 0.65) and rankinite (δ_{iso} = -74.5 and -75.9 ppm with Δ = -55.3 and -40.5 ppm and η = 0.69 and 0.65, respectively) [47,53]. Within the slag, these chemical environments can be interpreted as two types of Q^1 environments one in an Al-rich environment like in gehlenite and the other surrounded by alkaline modifiers like Ca and Mg like rankinite. The presence of other alkaline earth cations and other impurities increases the Δ of the rankinite Q^1 component and has an influence on the measured sign of Δ.

After the MCAA process, there is a dramatic increase of the CSA Δ, nearly doubling from 43 to 83 ppm, (Fig. 4b) despite minimal change in the isotropic spectrum in comparison to the slag. The corresponding trend of η resembles that in the slag although is shallower along the extremities. The increase in CSA of the MCAA powder is the result of greater mixing during milling and the formation of new disordered structures. One contribution is from the mixing of Na alkali from Na_2CO_3 additive into the aluminosilicate network to form disordered Q^2 environments inferred by the agreement of isotropic chemical shift and Δ of Na_2SiO_3 (δ_{iso}= -77.6 ppm, Δ = 78 ppm, η =0.52) [55] with that measured for the MCAA powder. However, EDS indicates there is only 1 Na ion for every 3 Si atoms implying the overall influence on the ^{29}Si nuclei will be minor if Na acts to substitute an alkaline cation on a pre-existing non-bridging oxygen site. Thus, there must be an additional contribution to the larger CSA in the MCAA powder. A potential explanation is the formation of a disordered melilite-based local structure with closer resemblance to akermanite than gehlenite; the two minerals form of a solid solution having the same structure the only difference is the cation on the T_1 site, Al for gehlenite and Mg for akermanite [57]. Notably the δ_{iso} for gehlenite and akermanite are similar (-72.5 and -73 ppm, respectively) however the CSA for akermanite, Δ=-72 ppm, is much larger than gehlenite, Δ=-48 ppm [58]. Prior studies investigating akermanite-gehlenite glasses reveal minimal effect of mixing on the ^{29}Si isotropic chemical shift [59] and explains why the isotropic line shape does not significantly change between the slag and the MCAA powder. The MCAA powder isotropic line shape can then be deconvoluted in a similar manner as the slag, where the lower frequency peak is composed of both gehlenite-like Q^1 environments as well as akermanite-like Q^1 environments having more Mg than Al and the higher frequency peak is composed of Q^1 and Q^2 environments with Na incorporated into the network.

After hydration, the CSA reverts to the smaller values observed in the slag with variation of the CSA parameters between -68 and -85 ppm following a similar trend as seen in the slag. This implies the reacted akermanite-like regions of the aluminosilicate network transform leaving the signal of the unreacted gehlenite-type regions, supported by the agreement of CSA values with gehlenite. Between δ_{iso} -74 and -81 ppm, Δ plateaus at 44 ppm and changes sign to negative, a strong indication that the CSA is from Q^1 units, likely of the rankinite-type. With decreasing frequency towards -90 ppm, Δ increases to 60 ppm and is tentatively assigned to bridging Q^2 environments in the presence of hydroxyl groups like those found in C-S-H gels. These sites have Δ values between 50 and 90 ppm [56] and their assignment is further supported by the CP signal centered at -87 ppm (Fig. 3a) implying these units must be associated with hydroxyl groups or structural water. Other contributions to this peak and another peak at lower frequency are various hydrous dreierketten-like Q^2 and Q^3 units found within xonotlite and tobermorite-type structures [53,56]. The combination of the overlapping signal from the Q^1 peak with negative Δ and the high η value of the Q^2 units causes the sign of Δ for the Q^2 units to be negative. These results indicate that only a portion of the aluminosilicate network is reacted by the MCAA process to form disordered sodium silicates and dimeric Mg-rich akermanite-type Q^1 units from the gehlenite-type slag. Upon hydration, the reacted akermanite-type regions condense to form a chain-like structures typical of those observed in C-N-A-S-H gels [41].

Together, the ^{27}Al and ^{29}Si NMR suggests the following structural picture of the MCAA process and hydration. The initial slag is composed of an amorphous melilite type aluminosilicate network of dimeric Q^1 Si and Al tetrahedra that are mostly cross-linked by Al and separated by regions rich in Ca and Mg cations. The MCAA process reacts a portion of the slag by increasing mixing of Mg into tetrahedral 'crosslinking' positions previously occupied by Al and then the Al is integrated into small fractions of Q^2 sodium silicate chains beginning to form. After hydration, these reacted melilite aluminosilicate regions condense by forming longer hydroxy-aluminosilicate chains composed of Q^2 and some Q^3 units. This condensation is driven by the expulsion of Ca and Na ions from the network to form metastable and stable carbonate phases (discussed further in the next section). Despite some similarities, the resulting gel microstructure of the MCAA process differs from the C-A-S-H gel of AAM by the absence of significant octahedral Al sites and the ^{27}Al and ^{29}Si spectra remain broad relative to the sharper features typically observed in cured C-A-S-H gels [14,60].

3.3 ^{13}C NMR Investigation of Carbonate Phases

Solid-state ^{13}C NMR can elucidate the carbonate phases and extent of carbonation reaction during hydration. Unfortunately, the resonances of many inorganic carbonates fall within a very narrow range (168.5-171.2ppm) [61] making exact phase identification challenging within these complex systems. To aid in phase identification the ^{13}C{^{1}H} CP/MAS experiment is used to selectively filter for hydrated carbonate phases. The single pulse (SP) and ^{13}C{^{1}H} CP/MAS NMR measurements for the MCAA powder and paste are shown in Figure 5.

There are striking differences between the line shapes of the two experiments for both the MCAA powder and paste, indicating the carbon atoms are segregated into hydrous and anhydrous phases. The SP line shape of the MCAA powder (Fig. 5a) has two broad peaks centered around 165 and 171 ppm with considerable intensity between them while the CP line shape is dominated by two overlapping components centered around 165 ppm and only has minor intensity at higher frequency. In conjunction with the bicarbonate phases observed in XRD, the major peaks in the CP spectrum can be assigned to nahcolite ($NaHCO_3$) based on similar isotropic chemical shift ($NaHCO_3$

δ_{iso}= 164.55 ppm, see Fig. S6) and the other tentatively to the mineral trona ($Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$), although the isotropic chemical shift has not been previously reported. The lower intensity peaks in the CP spectrum are associated with chemical environments farther away from protons and may not be directly associated with any hydrated carbonate phases [62].

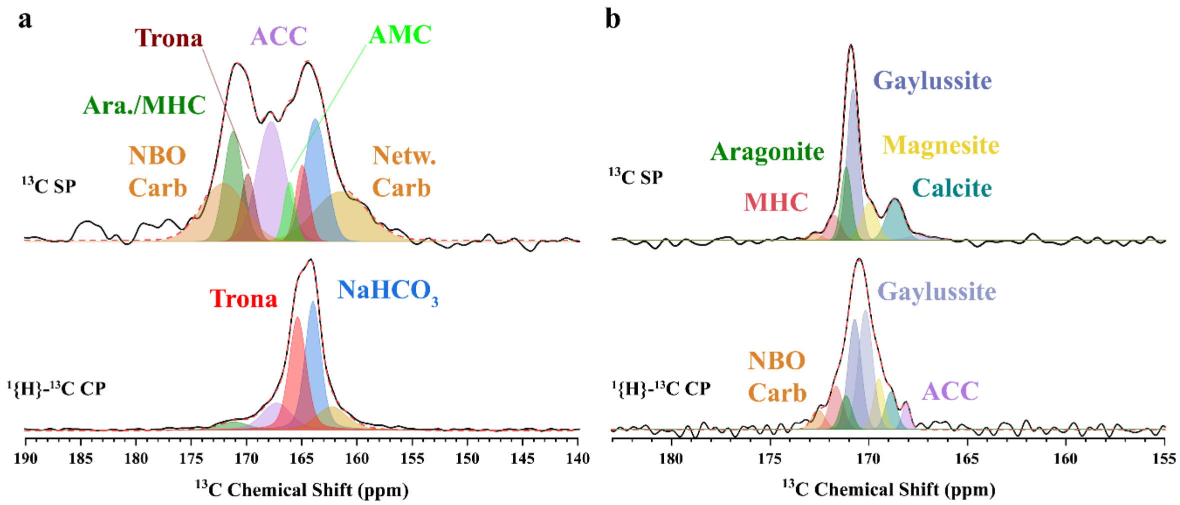

Figure 5 Single pulse (SP) and cross-polarization (CP) ^{13}C MAS NMR spectra of a. MCAA powder and b. MCAA paste. The deconvolution and corresponding assignment of the carbon phases are indicated by coloration of the peaks and summarized in Table S3.

As the right peak of the MCAA powder SP line shape is associated with bicarbonates the broad left peak is attributed to carbonate phases. However, precise identification is challenging due to the underlying structural disorder resulting in a distribution of chemical shifts as well as the incredibly narrow region carbonates occupy (< 3 ppm). Despite these setbacks there should be a corresponding sodium carbonate-like resonance from trona which we tentatively assign at 169.97 ppm. The peak at 171.29 ppm is assigned to a combination of aragonite (δ_{iso} =171.2 ppm) and monohydrocalcite (171.1 ppm) based on similarity to their respective chemical shifts [63], however the low intensity in the CP spectra points to a greater concentration of aragonite. The intensity in the center of the two broad peaks in the SP spectrum is assigned to thermodynamically metastable amorphous calcium carbonate (ACC, δ_{iso}=167.8 ppm), which has important implications for understanding the reaction pathway for forming stable carbonates in the MCAA process [63,64]. The small shoulder to the right of the ACC peak is attributed to amorphous magnesia calcite (δ_{iso}=166.3 ppm) [65].

The remaining intensity at the edges of the MCAA SP spectra (>171 ppm and <162 ppm) falls outside the range expected for typical alkaline carbonates and their broad line shapes point to considerable structural disorder. Intensity in these regions have been previously associated with carbonates directly incorporated within the aluminosilicate network, commonly seen in melilite (Ca and Mg rich) and nephelinite (Na rich) glass compositions [66–68]. These 'network carbonates' exist within complexes as either Si-CO_3-Al linkages where CO_3 shares two bridging O with two neighboring Si/Al tetrahedra, or as molecular CO_2 weakly bonded to a bridging O between two corner-sharing tetrahedra. The peak at 161.55 ppm is assigned to these types of network carbonates. The peak at 172.19 ppm is assigned to a different type of carbonate incorporated into the aluminosilicate network, 'non-bridging carbonates' (NBO carbonate). These are carbonates associated with non-bridging oxygens in the aluminosilicate network that are partially or completely charge balanced by nearby sodium ions [69,70]. The ^{13}C spectra of the MCAA process at this stage indicates that the sodium carbonate additive and CO_2 rich atmosphere during milling is mostly transformed into bicarbonates, metastable amorphous carbonate phases, and aluminosilicate network carbonates.

After hydration and curing, the ^{13}C spectra of the paste (Fig. 5b) are significantly narrowed in comparison to the MCAA powder and lack intensity in the bicarbonate region. This narrowing indicates the overall transformation of the carbonates from amorphous to crystalline, corroborated by the well-defined reflections of calcite, aragonite, and gaylussite in the XRD pattern and their presence aids in deconvolution of the SP and CP spectra. In the SP spectrum, the calcite peak (δ_{iso}=168.8ppm) is clearly distinguishable while aragonite and gaylussite are overlapping due to similar chemical shifts (Fig. S6) although deconvolution indicates gaylussite is the major carbonate species formed (Table S3). Secondary peaks are identified by shoulders in the line shape as magnesite and monohydrocalcite. Residual intensity from ACC and the NBO carbonate can be seen, indicating the carbonates have nearly entirely precipitated out of the aluminosilicate network to form more stable phases. The CP spectrum can be deconvoluted with the same species and indicates nearly all the carbonate environments are in some proximity to water molecules, while gaylussite shows the highest intensity due to its structurally bound water molecules. Overall, the ^{13}C NMR indicates that upon hydration there is substantial chemical and structural transformation of the carbon environments with nearly complete dissolution of sodium bicarbonate. This dissolution supplies CO_3^{2-} for carbonation reactions with Ca/Mg^{2+} to form anhydrous $CaCO_3$ phases, aragonite and calcite, and $MgCO_3$, magnesite, and hydrous gaylussite and monohydrocalcite. A secondary pathway for forming stable carbonate phases is inferred from the absence of the ACC peak, suggesting that with curing the metastable ACC crystallizes directly into calcite or aragonite.

3.4 FTIR Spectroscopy

The FTIR transmittance spectra (Fig. 6) contain many broad and overlapping contributions that can be interpreted by the phase assignments made from XRD and NMR results. The spectrum of the slag bears a striking resemblance to the FTIR spectrum for gehlenite and akermanite glass mixtures [59]. The spectra of gehlenite/akermanite mixtures including the slag presented here, display a broad asymmetric band from 800 to 1200 cm^{-1} with the center of gravity around ~950 cm^{-1}, this region is attributed to stretching modes of small chain Q^2 and dimeric Q^1 Si-O-Si/Al linkages [59]. Bands between 600-450 cm^{-1} are associated with the deformation vibrational modes of various tetrahedral species like Si, Al, and Mg that could be expected for melilite type silicate glasses and bending modes of Si-O-Si/Al linkages [35,39,59,71]. The broad band intensity at 695 cm^{-1} is from the stretching mode of tetrahedral Al-O, in agreement with the observation of tetrahedral Al via ^{27}Al NMR.

After the MCAA process, sharp bands appear on top of the broad bands associated with the melilite glass due to the formation of various carbonates and bicarbonates previously identified with ^{13}C NMR. Sodium bicarbonate is clearly observed by the well-defined stretching mode at 1923 cm^{-1} [72].

Formation of carbonates is apparent by the broad band at 1480 cm^{-1} associated with the asymmetric stretching of C-O in carbonates [39], the ν_4 mode of CO_3^{2-} at 705 cm^{-1} [73,74] and the broad bands at 2552 and 2915 cm^{-1} associated with ν_1 + ν_3 and 2ν_3 modes of CO_3, respectively [75,76]. As observed in the ^{13}C NMR, a portion of the carbonate molecules are integrated into the aluminosilicate network giving rise to the broad band at 1358 cm^{-1} for stretching modes of bridging CO_3^{2-} between tetrahedral Si and Al (expected range 1345-1386 cm^{-1}) and the shoulder band 1290 cm^{-1} for stretching of non-bridging CO_3^{2-} in the aluminosilicate chains (expected range 1250-1290 cm^{-1}) [68]. The relatively sharp bands at 1665 and 1623 cm^{-1} are from deformation modes of free and chemically bonded water, respectively; the rise of a broad band at 3435 cm^{-1} is associated with O-H stretching of hydroxide in carbonates [73,74,77]. While molecular water is observed, the lack of signal from the cross-polarization ^{29}Si NMR measurement for the MCAA powder indicates these structurally bound water molecules are not associated with the aluminosilicate structure, rather they are almost exclusively associated with bicarbonates (sodium bicarbonate and trona).

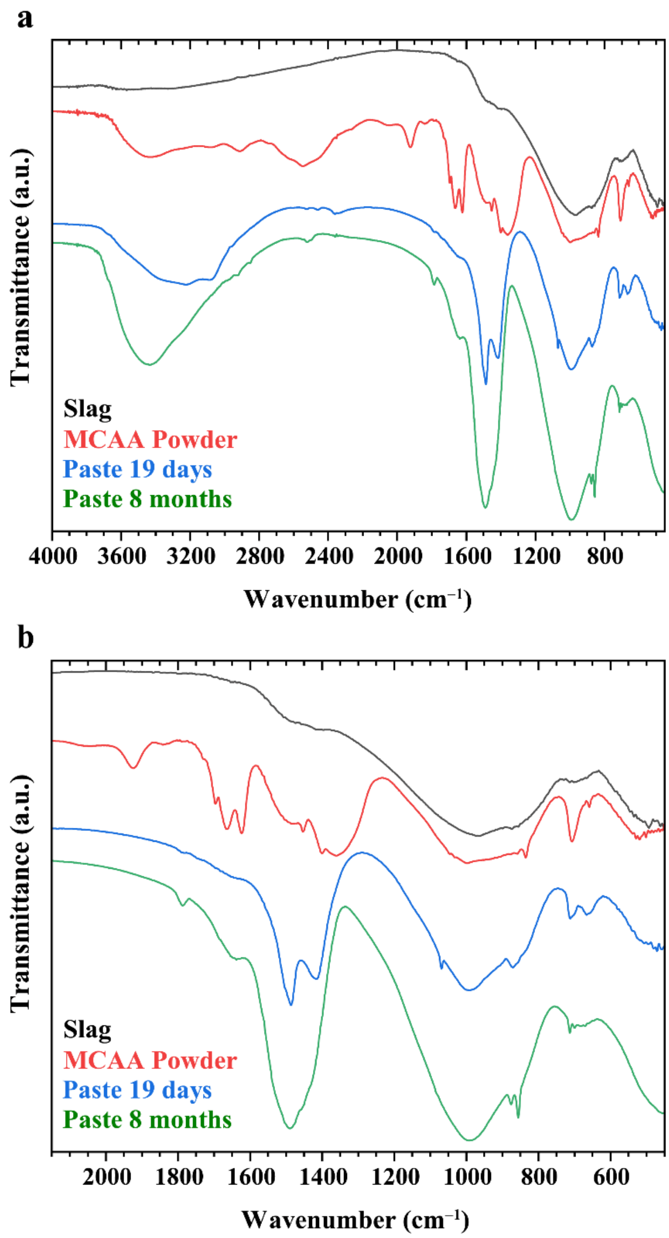

Figure 6 FTIR spectra of the slag, MCAA powder, and paste after different curing durations. The full spectra range is provided by a. with a narrowed section focusing on the carbonate and aluminosilicate region provided by b.

After hydration, the band at ~1660 cm^{-1} associated with chemically bonded water, the band for sodium bicarbonate (1923 cm^{-1}), and the aluminosilicate carbonate bands between 1260-1350 cm^{-1} nearly disappear as the intensity of carbonate stretching modes at 1487, 1420, and 871 cm^{-1} greatly increase. This implies the bicarbonate species transform during the initial hydration into stable carbonates, completely consistent with ^{13}C NMR. Previous studies on alkali activation of slag show structurally bound water is consumed on a timescale of tens of hours after hydration, thus it is not surprising the bands for bound and free water are absent in the present spectrum [77]. The stable carbonates anticipated from XRD and ^{13}C NMR (calcite, aragonite, gaylussite) all have similar stretching modes in the 1400 to 1490 cm^{-1} region making individual assignment difficult [78]. The polymerization of the aluminosilicate network is noted by the increasing intensity at 994 cm^{-1} which is attributed to stretching of Q^2 units in C-N-A-S-H type gels [77]. With prolonged curing towards 8 months, the intensity of the bands at 994 and 1490 cm^{-1} greatly increases indicating continued polymerization of the aluminosilicate network into a C-N-A-S-H type gel and formation of stable carbonates with hydration time. Further insight that the microstructure evolves towards a C-N-A-S-H type gel is the reappearance of the band for chemically bound water at 1643 cm^{-1}.

3.5 Thermogravimetric analysis

Thermal stability of the cementitious materials and the constituent carbonate phases can be inferred from TGA (Fig. 7a) by measuring the thermal decomposition and evolved gas as a function of temperature. The precursor slag shows very little weight loss (less than 1 wt%) even up to 900°C whereas the MCAA powder and paste lose an accumulated 10 to 20 wt% in the same temperature range. The derivative TGA curve (Fig. 7b) provides insight into when thermal decomposition and out-gassing begins and aids in identification of the constituent products. Again, the slag shows no significant changes while the MCAA powder and paste have two discernable decomposition regions, a sharp decomposition around 120°C and broader decomposition from 550 to 700°C.

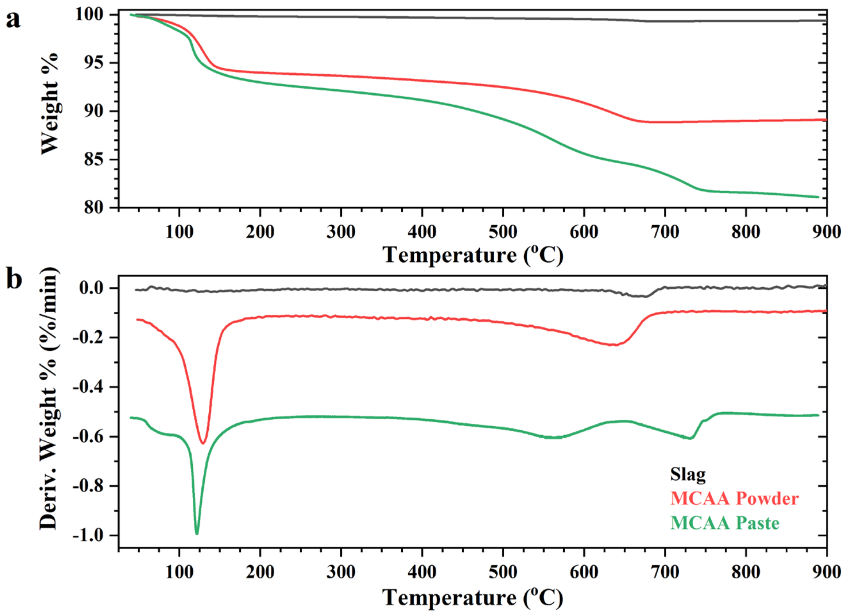

Figure 7 a. TGA curves of the slag, MCAA powder, and paste b. Differential TGA curves of the same samples; curves are offset for clarity.

These are both attributed to the newly formed (bi)carbonate phases and sodium carbonate additive and not affiliated with the underlying aluminosilicate network as seen in the slag derivative TGA curve. Within the first region, the peak max decreases marginally from 129°C for the MCAA powder to 123°C in the paste as well as sharpens. Potential thermal decomposition pathways in the MCAA system are complicated by its multiphase nature and their morphological interactions thus discerning specific peaks to a sole decomposition product is unlikely. However, the structural model presented can be used to interpret the thermal decomposition products. The first peak of the MCAA powder is attributed mostly to the direct dehydration and decarbonation of sodium bicarbonate which by itself decomposes [79] to produce H_2O and CO_2 in the range of 100 – 180°C and trona which has a two-step decomposition to produce H_2O, sodium carbonate, and sodium bicarbonate at ~109°C followed quickly by the decomposition of sodium bicarbonate to produce CO_2 at ~123°C [80].

In the hydrated paste, the sharp peak is shifted slightly to lower temperature and is preceded by a broad component. This broad component is attributed to the dehydration of adsorbed water, while the sharp peak is attributed to the dehydration of gaylussite into sodium calcium double carbonate and H_2O [38]. The dehydration of gaylussite is not accompanied by any release of CO_2 as shown in section 3.6. At higher temperatures, the MCAA powder and paste display different thermal events between 560°C to 750°C corresponding to decarbonation of carbonate phases. The breadth of the peaks makes exact assignment difficult however lower temperature (550-700°C) decomposition of calcium carbonate can occur if it is a fine grain microstructure or as a relatively unstable phase like aragonite [81]. In the paste, a similar explanation can account for the broad peak around 560°C while the 730°C peak is attributed to the decomposition of the sodium calcium double carbonate from gaylussite into sodium carbonate, calcium oxide, and CO_2 [38].

3.6 Variable Temperature FTIR of Evolved Gas Products

To supplement the identity of the decomposition products inferred by DTGA, variable temperature FTIR of the gaseous products is performed to obtain a better understanding of when the cementitious materials are decarbonating. The spectrum of the evolved gaseous product clearly shows CO_2, given by the doublet at 2340 and 2360 cm^{-1} for the CO_2 stretching mode (Fig. 8). With a spectral resolution of 2 cm^{-1}, we can integrate each spectrum between 2300 cm^{-1} and 2380 cm^{-1} to accurately determine the variation of CO_2 concentration as a function of temperature, which is plotted in Figure 8b. The temperature dependent CO_2 concentration agrees with the DTGA, showing no CO_2 production in the granulated slag, a significant quantity is produced after 100°C in the MCAA powder, and begins to be released around 300°C in the paste. The CO_2 release at low temperatures in the MCAA powder supports the assignment that this decomposition is from bicarbonates. The lack of CO_2 production until 300°C in the paste also supports the previous assignment that stable hydrous carbonate phases like gaylussite are formed and only thermal dehydration occurs below 300°C.

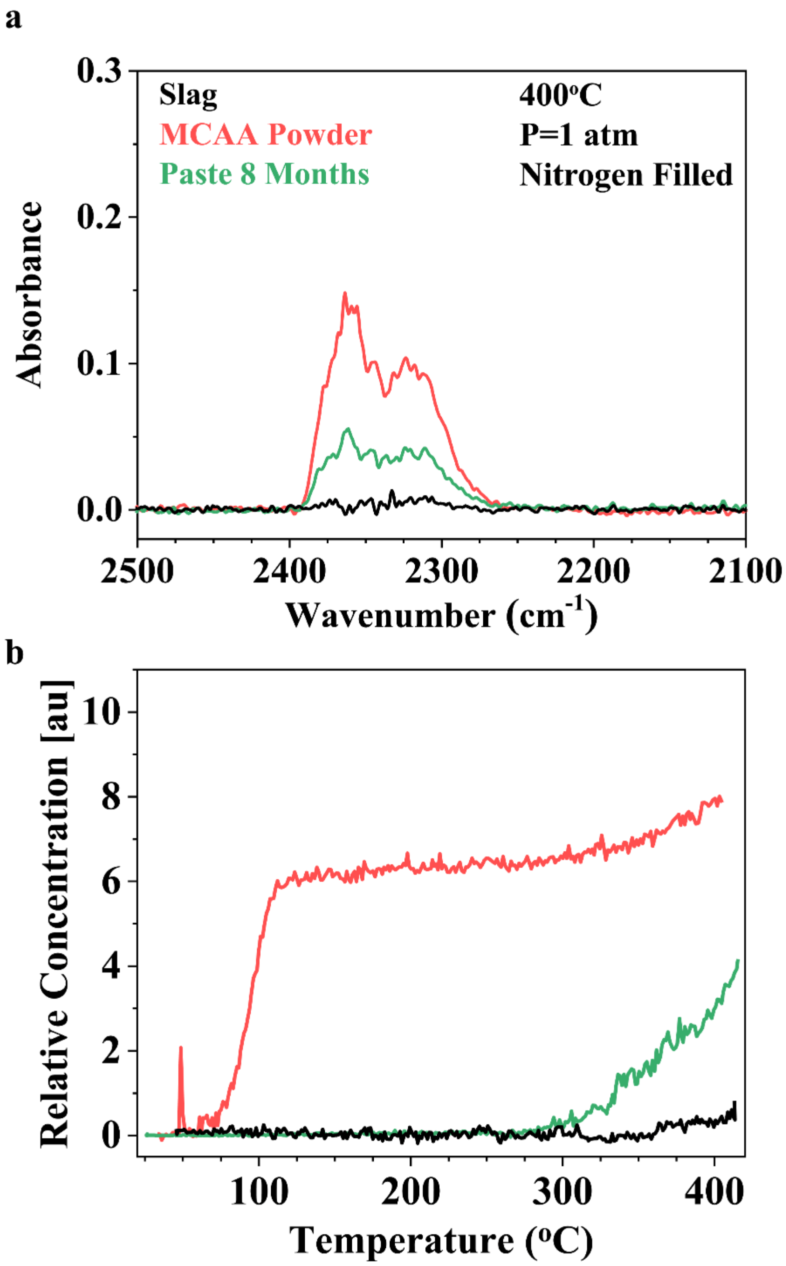

Figure 8 a. Representative IR spectra of the evolved CO_2 gas from thermal decomposition at 400°C. b. The relative concentration of evolved CO_2 gas from thermal decomposition of the slag, MCAA powder and paste.

3.7 Total Inorganic Carbon

Finally, to quantify the amount of carbon captured during the MCAA process total inorganic carbon testing on the powder is performed. The TIC measurements show the total CO_2 content is 10.4 ± 1.3 wt% in the MCAA powder, after subtracting the 4 wt% CO_2 expected from the 10 wt% addition of Na_2CO_3, the total captured CO_2 from the milling process is 6.3 ± 1.3 wt%. This result is rather remarkable as it shows that the MCAA process can be used as a form of room temperature capture of CO_2 while upcycling alkaline waste products to form cementitious materials.

3.8 Accelerated Carbonation through Mechanochemical Milling and Alkali Activation

Traditional aqueous carbonation is known to occur in curing cement by way of a two-step mechanism wherein CO_2 first dissolves into the pore solution from ambient air and reacts with H_2O to form carbonic acid (H_2CO_3). The carbonic acid then reacts with the available alkali and alkaline cations in the pore solution to form carbonates [11]. The sodium and potassium carbonates have high solubility in water and reversibly dissociate. On the other hand, the reaction with calcium is irreversible thus stable calcium carbonate progressively builds up in the pore capillaries. With continued precipitation of calcium carbonate, Ca ion transport to the pore solution is greatly diminished by the limited Ca diffusivity through calcium carbonate and the Ca depleted silicate zone [18,19].

With this two-step mechanism in mind, the extent of the carbonation reactions is limited by either diffusion of CO_2 into the pore solution or Ca ion diffusion through the anhydrous silicate and passivation carbonate layers. Thus, many prior accelerated carbonation techniques are centered around increasing the reaction kinetics by increasing Ca ion and CO_2 diffusivities through use of high CO_2 partial pressures or curing at elevated temperatures [6–9]. These methods for accelerating carbonation reactions, along with the carbonation in AAM, occur during the hydration step of cement production, while in comparison the MCAA process has most of the carbonation occurring before hydration as the CO_2 is reacted and captured in metastable bicarbonate and amorphous carbonate phases.

Given what is known about aqueous carbonation mechanisms and the carbonate phases observed in this study, the MCAA process likely follows the same carbonation reaction steps however the extent of carbonation is rapidly accelerated by a synergistic combination of milling and high concentration of active sodium ions in a CO_2 rich atmosphere. From a microstructural picture of carbonation, carbonic acid is deposited on the pore surface as bicarbonate and builds up as a passivation layer preventing Ca from leaching into the pore solution and limits Ca transport to the slag/bicarbonate interface. The effect of milling breaks off this passivation layer to supply unreacted surface and allow the reaction to continue. Mechanochemical milling bypasses the slow kinetics imposed by diffusion and allows the carbonation reaction to occur at a much faster rate at the newly exposed surface.

The addition of sodium carbonate and milling in a CO_2 rich atmosphere are also key. Sodium carbonate acts as a buffer for the pore solution by keeping the pH high and readily reacts with gaseous CO_2 to supply a continuous source of sodium bicarbonate that then reacts with the slag surface. While this bicarbonate formation mechanism accounts for most of the carbonation, there is evidence of a secondary carbonation mechanism by trapping of the carbonate molecule within the aluminosilicate network and forming amorphous calcium carbonate. The results from ^{13}C NMR and FTIR also suggests some carbonates are incorporated into the aluminosilicate network as both bridging and non-bridging complex species between Si and Al tetrahedra. The nature of these 'network carbonates' can be elucidated by a study on binary sodium silicate glasses that show carbonate molecules are surrounded by Na ions to form nanoscale domains within the silicate network and increase proportionally with the degree of network depolymerization. It was shown for highly modified glasses (> 50 mol% Na_2O) the CO_2 retention in the glass as carbonate is greater than 0.5 wt% [82]. This observation indicates that the alkaline/alkali-rich aluminosilicate network of the precursor slag used in this study can contain a non-trivial amount of CO_2 as carbonate molecules within the aluminosilicate network. The extreme and localized pressure and temperature fluctuations that occur during milling likely give rise to this direct mixing of the carbonate molecules within the aluminosilicate network. These extreme conditions can also explain the observation of amorphous calcium carbonate in the ^{13}C NMR spectra, as the intense mechanical energy leads to amorphization of the particulates including the exfoliated bicarbonate passivation layers. Some of these bicarbonate passivation layers are supplied with Ca ions during the intense mixing to form calcium carbonate.

The carbonation phases resulting from the MCAA process, i.e., bicarbonates, network carbonates, and ACC, are metastable as indicated by their continued presence in the sample after 8 months. The origin of CO_2 to form the carbonate phases comes from both CO_2 gas during milling and from the sodium carbonate additive. It is unclear at this time which specific carbonate phases can be sourced to CO_2 gas and the additive. After hydration, rapid crystallization and phase transformations occur to form the calcium carbonate phases calcite and aragonite and hydrated carbonate phase gaylussite. It is likely that the metastable sodium bicarbonate undergoes a hydraulic reaction with Ca ions in the pore solution to produce gaylussite as these two phases make up a large fraction of their respective ^{13}C NMR spectra. The leaching of Ca ions from the aluminosilicate network could drive the network carbonates to precipitate out of the glass network as separate carbonate phases, suggested by the absence of these signals in the ^{13}C NMR and FTIR spectra of the cured paste. Hydration also appears to cause the ACC phase to crystallize into calcite and aragonite potentially by a seeding mechanism from one of the newly nucleated carbonates. The result of these radical changes from metastable to stable carbonates leads to needle-like protrusions penetrating throughout the glassy material as observed in SEM. The morphology and extent of reaction observed for these calcium carbonate phases in the aluminosilicate material is promising for its application as a cement binder.

Previously, similar experiments investigating the effect of mechanochemically milling GBFS:lime mixtures in CO_2 rich atmospheres show a positive increase in the compressive strength with 28 day values of ~27 MPa, noting a similar evolution of the carbonate phases binding the microstructure [20]. These prior studies also suggest that the formation of gaylussite only occurs when the slag is milled in a CO_2 rich atmosphere. This supports the proposed theory that the alkali additive acts as a buffer to maintain a high pH and the reaction is sustained to a much further degree as new surface is exposed during the milling process. Parallel to the microstructural evolution of the carbonates, the aluminosilicate network undergoes polymerization as Ca ions are depleted during hydration to form longer chains. While the aluminosilicate network remains largely amorphous, the local chain structure bears some resemblance to those seen in C-N-A-S-H gels where Al is incorporated in the chain backbone as a tetrahedral species [16,17]. The rather dramatic hydraulic reactions to form a C-

N-A-S-H type gel indicate the efficacy of the MCAA process to upcycle a hazardous waste into a cementitious material that can be used either by itself or as an additive to cement blends for reducing CO_2 emissions during cement production.

4. Conclusions

Using multi-analytical characterization techniques, the MCAA process is found to have profound effects on transforming a low-value material, slag, into one that shows extensive carbonation and has the potential for hydraulic reactions. The combination of alkali activation and mechanochemical milling surmounts the diffusion limited carbonation reaction and creates the conditions to rapidly accelerate carbonation as the alkali additive buffers the pH and the milling continually creates unreacted surfaces for carbonation to occur. This process is unique from other processes for creating cementitious materials with accelerated carbonation because the MCAA process creates a clinker material with advanced carbonation before hydration and captures CO_2 from the CO_2 rich atmosphere in the form of metastable carbonates during production.

There are many other engineering considerations like pH and mechanical strength that need to be investigated before utilizing this MCAA process in technologically relevant cementitious applications. However, the MCAA process is a promising direction for decreasing the carbon footprint of cement production and utilizing previous alkaline waste materials that would otherwise pollute the environment. The MCAA process for producing clinker material shows that accelerated carbonation through increased kinetics by exposing unreacted surface can be done in a single, relatively low energy step. It is a particularly appealing as the process would be relatively easy to integrate into current cement manufacturing procedures.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary Information

Electronic Supplementary Information contents: Details of variable temperature IR methods, 2D contour plots for ^{27}Al 3QMAS and ^{29}Si MATPASS NMR experiments and analysis, ^{13}C MAS NMR spectra of reference carbonate minerals gaylussite, Na_2CO_3, and $NaHCO_3$.

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